

ALDAN, Russian Federation

WMO: 310040

Lat: 58.617N

Lon: 125.367E

Elev: 2228

StdP: 13.55

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.3	-36.8	-43.9	0.5	-39.8	-40.7	0.6	-36.7	14.2	2.3	12.2	-0.2	1.0	200	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	82.1	63.4	78.5	62.3	74.9	60.5	66.0	77.0	64.2	74.6	62.4	72.2	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	91.5	70.0	60.3	84.8	68.3	58.4	79.2	67.4	32.1	76.8	30.7	74.5	29.3	72.5	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.6	9.3	DB	-43.4	87.1	4.5	2.9	-46.7	89.2	-49.4	90.9	-51.9	92.5	-55.2	94.6	(4)
(5)			WB	-43.5	68.5	4.5	1.7	-46.7	69.7	-49.3	70.7	-51.8	71.7	-55.1	73.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.3	-15.6	-9.8	4.8	24.2	41.5	57.3	62.2	56.6	41.0	20.6	-2.2	-15.1
	DBStd	30.01	12.74	11.30	12.19	9.66	8.13	7.99	7.07	7.71	8.22	10.58	13.75	12.71
	HDD50	11007	2035	1675	1402	773	283	25	4	28	288	910	1565	2019
	HDD65	15670	2500	2095	1867	1223	727	250	138	276	720	1375	2015	2484
	CDD50	897	0	0	0	0	21	244	382	232	18	0	0	0
	CDD65	86	0	0	0	0	0	19	51	16	0	0	0	0
	CDH74	940	0	0	0	0	4	277	510	149	0	0	0	0
	CDH80	196	0	0	0	0	0	58	117	21	0	0	0	0
Wind	WSAvg	3.8	3.4	3.6	4.0	4.5	4.6	3.7	3.3	3.3	3.7	3.9	3.8	3.2
Precipitation	PrecAvg	26.7	1.0	0.9	1.1	1.5	2.6	3.3	3.8	4.2	3.5	2.5	1.5	1.1
	PrecMax	35.8	2.0	2.1	2.1	2.7	6.2	6.1	9.7	10.6	6.6	6.1	3.9	2.5
	PrecMin	18.4	0.2	0.2	0.3	0.5	0.8	0.2	0.0	0.6	0.8	0.8	0.5	0.4
	PrecStd	4.7	0.4	0.4	0.4	0.6	1.2	1.4	2.2	2.2	1.3	0.9	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	22.0	37.3	53.0	72.7	85.7	87.6	82.9	69.8	48.9	33.5	21.2
		MCWB	13.6	19.1	29.3	40.1	52.7	62.9	66.0	63.9	55.7	39.7	29.3	20.2
	2%	DB	10.6	16.7	30.6	46.7	65.5	80.9	83.2	78.4	63.7	42.2	28.0	12.6
		MCWB	9.2	14.7	25.5	36.8	49.5	61.1	64.5	62.8	52.2	36.6	25.8	11.3
	5%	DB	6.7	12.2	26.7	42.5	60.7	76.9	79.7	74.1	58.9	37.7	23.5	8.6
		MCWB	5.7	10.9	23.0	34.1	47.0	59.2	64.0	61.1	49.1	33.8	21.7	7.6
	10%	DB	2.9	7.7	23.2	38.7	56.4	73.1	76.1	70.4	54.5	34.0	18.4	4.3
		MCWB	1.9	6.5	20.4	31.7	44.7	57.2	62.6	59.7	47.4	30.9	16.7	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	19.2	30.2	40.9	54.2	65.6	68.4	67.5	57.2	40.9	30.3	19.5
		MCDB	14.7	22.0	35.2	50.5	69.7	79.7	80.0	76.5	66.1	46.3	32.9	22.2
	2%	WB	9.4	14.4	26.1	37.6	50.7	62.6	66.7	64.6	53.4	37.2	25.7	11.4
		MCDB	10.5	16.1	30.2	46.0	63.0	76.9	78.1	73.7	60.4	41.4	28.0	12.5
	5%	WB	5.8	10.8	23.2	34.7	48.1	60.6	65.4	62.6	50.7	34.0	21.7	7.7
		MCDB	6.6	12.2	26.4	41.2	58.6	74.0	76.2	71.9	56.8	37.5	23.4	8.7
	10%	WB	2.1	6.5	20.3	32.2	45.5	58.8	63.8	60.6	48.0	31.2	16.7	3.5
		MCDB	2.9	7.7	23.3	38.2	54.8	69.9	73.7	69.0	53.6	33.7	18.4	4.3
Mean Daily Temperature Range	5% DB	MDBR	13.5	16.3	18.9	17.9	16.6	20.5	18.5	17.7	14.0	11.9	13.0	12.2
		MCDBR	14.6	17.2	19.1	19.7	25.2	26.9	24.8	24.2	22.8	14.9	13.9	12.5
		MCWBR	13.9	15.8	15.5	12.3	12.9	11.8	10.6	11.2	12.8	10.2	12.4	12.0
	5% WB	MCDBR	14.5	16.8	17.5	18.0	22.7	23.1	20.3	20.5	19.5	13.5	13.6	12.3
MCWBR		13.9	15.4	14.5	11.7	12.1	11.3	9.9	10.4	11.8	9.7	12.3	11.9	
Clear-Sky Solar Irradiance	taub	0.234	0.265	0.295	0.364	0.436	0.409	0.454	0.405	0.331	0.285	0.239	0.216	
	taud	2.021	2.066	2.070	1.967	2.009	2.270	2.146	2.278	2.440	2.273	2.042	1.950	
	Ebn at Noon	184	238	266	262	251	262	245	250	257	232	178	146	
	Edh at Noon	16	27	36	48	49	39	43	35	25	22	16	12	
All-Sky Solar Radiation	RadAvg	207	544	1097	1639	1590	1753	1591	1245	802	422	259	129	
	RadStd	12	24	44	77	131	127	158	95	63	40	24	6	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.51	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	+1.37	N/A	N/A	N/A	N/A	

Nomenclature: See separate page